

DESIGNING INDIA'S CHIP DESTINY

AMD's Jaya Jagadish
on future-proofing
India in the world's most
complex industry

Jayesh Shinde
editor@digit.in

**JAYA
JAGADISH**
Country Head & SVP
Design Engineering,
AMD India

If you want to understand where India's semiconductor ambition stops being bravado and starts becoming silicon reality, you talk to Jaya Jagadish, Country Head & SVP Design Engineering, AMD India. For nearly two decades, she has helped turn AMD's India operations from a promising outpost into a 10,000-plus engineer design powerhouse that contributes to global CPU roadmaps across AI, gaming, and high-performance computing. But today, her brief extends far beyond a single company's P&L. As Chairperson of the Semicon Talent Building Committee under the India Semiconductor Mission, she helped craft the *Semicon India FutureSkills Talent Roadmap* – a blueprint meant to future-proof India's place in the world's most complex industry.

Jagadish remains clear-eyed about what India must measure

over the next five years: not just incentives announced or fabs inaugurated, but "how many products are actually designed and built in India for the Indian market." Edited excerpts from my interview follow:

What's the one semiconductor capability India must nail first?

Jaya: In today's complex and globally distributed semiconductor value chains, chip development has evolved into a highly collaborative effort spread across geographies. This increasing interdependence is not a limitation, rather it is the operating model of the modern semiconductor industry.

India already has a strong and well-established design presence, but the priority now will be to deepen that capability by enabling more indigenous startups to design complex system-level chips. Design

complexity is what will ultimately drive innovation and long-term value in semiconductors.

As the industry moves toward chiplet-based architectures and heterogeneous integration, advanced packaging will become essential. Packaging capabilities need to evolve alongside semiconductor design, especially as systems grow more complex. For advanced designs to succeed, design depth and advanced packaging must progress together, reinforcing each other as part of the same technology stack.

Meanwhile manufacturing has to get off the ground quickly and pragmatically. We need to start with building the right skills, attract investment, and align manufacturing with designs being developed locally. India will see progress by strengthening all three pillars in